

RSMA-Based Visible Light Covert Communications With Jammer Assistance for UAV Systems: Joint Trajectory Design and Resource Allocation

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Abstract—Uncrewed aerial vehicles (UAVs) enable flexible network deployment but are vulnerable to detection due to the open-access nature of wireless signals. Visible light communication (VLC) emerges as a compelling technology for this context, offering inherent covertness through its reliance on line-of-sight links, while its wide spectrum supports high covert transmission rates. However, the high mobility of UAVs creates dynamic channel conditions that challenge traditional multiple access schemes. This motivates the utilization of rate-splitting multiple access (RSMA), a more robust and flexible framework for managing multi-user interference. Therefore, we investigate a cooperative, RSMA-based VLC system where a source UAV provides covert data transmission, assisted by a jamming UAV that creates power randomness to confuse wardens. By jointly optimizing the three-dimensional (3D) trajectory design and resource allocation of UAVs, we aim to maximize the minimum average covert transmission rate of users. To achieve this goal, the original problem is decomposed into three subproblems, namely horizontal coordinate arrangement, altitude adjustment, along with joint power assignment and rate-splitting, which are solved by the successive convex approximation method, geometric programming-based approach, and majorization-minimization algorithm, respectively. Numerical results validate the effectiveness of the proposed approach through extensive simulations under various parameters and comparative analyses against baselines, highlighting the potential of UAV-assisted visible light covert communications.

Index Terms—Uncrewed aerial vehicle (UAV), covert communication, visible light communication (VLC), rate-splitting multiple access (RSMA), minimum average covert transmission rate, trajectory design, resource allocation.

I. INTRODUCTION

UNCREWED aerial vehicles (UAVs) offer significant advantages over traditional base stations (BSs) due to their high mobility, flexible deployment, and ability to provide on-demand coverage [2], [3]. These features make UAV-assisted networks well-suited for various applications, such as disaster recovery, emergency communications, and surveillance [4]. However, the open-access nature of wireless communications exposes UAV-assisted networks to security threats, as wardens may intercept confidential data transmissions [5]. To address these risks, covert communications have emerged as an effective technique, aiming to conceal the transmission behavior by minimizing the detection probability of wardens [6]. In covert communication scenarios, UAVs can undertake various roles, including serving as a BS [5], receiver [7], relay [8], and jammer [9]. To effectively leverage UAVs in these diverse roles, it is essential to consider both the limitations and benefits posed by their unique spatial and temporal degrees of freedom.

The inherent mobility of UAVs leads to dynamic channel gains and interference, posing challenges to traditional multiple access (MA) schemes [3]. To address this issue, recent studies have introduced rate-splitting multiple access (RSMA) into UAV systems, demonstrating its advantages in terms of the throughput [10], energy efficiency [11], and spectral efficiency [12]. The core principle of RSMA involves splitting messages into common and private parts, where the common part is decoded by multiple users, while the private part is decoded individually. Therefore, RSMA is able to balance between fully decoding interference and treating it as noise, depending on the interference level [13]. By leveraging this flexibility in interference management, RSMA provides strong robustness against the imperfect channel state information (CSI) in mobile UAV scenarios, mitigates the low signal-to-interference-plus-noise ratio (SINR) commonly encountered in covert communications, and minimizes energy leakage to unauthorized receivers [6], [14]. Consequently, under the same covertness requirement, RSMA can achieve a higher covert transmission rate (CTR) than traditional MA schemes [15].

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Besides, the joint optimization of UAV position and resource allocation can enhance covert channel gains while suppressing the reception of wardens [9]. For example, the authors in [16] analyze the flight region by partitioning it and employ a search-based method to identify the solution that maximizes the user's signal-to-noise ratio (SNR). For covert communications without direct links, literature [17] introduces a UAV-mounted reconfigurable intelligent surface framework to facilitate signal relaying, aiming to improve the CTR. However, these studies consider simplified scenarios involving a single user and a single warden. To extend coverage to multiple users over a larger area, the trajectory design becomes essential. In [3], the average CTR is maximized in a non-orthogonal multiple access (NOMA)-based uplink message collection scenario by jointly optimizing the UAV trajectory and transmit power. Moreover, the authors in [18] propose an energy-efficient covert transmission scheme that accounts for the propulsion energy and acceleration limits of UAVs, aiming to boost the minimum average CTR (MACTR). Additionally, by characterizing UAV trajectory as a successive-hover-and-fly structure, the fairness among users is improved by the sequential convex programming method [19]. Compared with the above works, which mainly focus on horizontal trajectory design of UAVs, the exploitation of three-dimensional (3D) mobility offers new opportunities for covert communications. For instance, a UAV-aided short-packet transmission scenario is studied in [20], focusing on the joint optimization of 3D trajectory and communication resources in the orthogonal multiple access (OMA) scheme. Moreover, to defend against active wardens that interfere with legitimate users and adopt collusive detection strategies, a robust algorithm incorporating 3D trajectory planning and transmit power control is developed [21].

From the perspective of randomness, the prior studies exploit the channel fading [3], [7], jamming power [9], [24], environmental noise [19], [22], and Gaussian signaling [21], [23] to enhance the covertness. Among them, jamming is particularly advantageous because it can be deliberately manipulated [2]. For instance, directional jamming targeting the warden is explored in [27]. In some cases, the jamming signal is known to legitimate users and can be canceled from the received signal [28], thereby further reducing interference to them. Moreover, even in the absence of coordination, the uninformed jammer is shown to guarantee a positive covert rate regardless of the warden's receiver type [29]. Existing research on UAV-assisted covert communications has investigated various jamming strategies. For instance, the authors in [7] employ stationary multi-antenna ground jammer, while literature [24] considers the single UAV equipped with two antennas for simultaneous data collection and jamming. More recent studies focus on a dedicated jamming UAV cooperating with UAV-mounted BSs, aiming to degrade the detection probabilities of wardens while minimizing the interference to legitimate users [9], [18].

However, most existing works focus on the radio frequency (RF)-based UAVs, which may interfere with terrestrial networks and exacerbate the spectrum congestion [30]. As an effective complementary technology to traditional RF com-

munications, visible light communication (VLC) employs light-emitting diode (LED) for transmission and photodiode (PD) for reception [31]. Since optical signals do not interfere with RF signals, VLC-enabled UAVs present inherent compatibility with existing ground networks [30]. Moreover, the mobility of UAVs allows dynamic altitude adjustment to alleviate signal blockage problems in VLC links, enhancing the communication reliability [32]. Therefore, integrating VLC into UAVs presents a viable alternative to mitigate the limitations of RF-based counterparts. For covert communications, VLC also provides natural advantages over RF since optical signals rely on line-of-sight (LoS) links and cannot penetrate obstacles, significantly reducing detection risks [33]. Furthermore, the sufficient spectrum of VLC maintains a high CTR, ensuring both covert and efficient communications [25], [26].

While works in RF systems offer valuable insights, the methods cannot be directly applied to VLC due to the distinct signal and channel models. Specifically, optical signals are subject to non-negative and real-valued constraints. When the peak-to-average optical intensity ratio takes on a certain value, the optimal signal intensity distribution is nearly uniform [34], which differs from the Gaussian distribution commonly assumed in covert RF communications. Moreover, the channel gain of VLC depends on distance, irradiance angle, and incidence angle [35], which are all influenced by UAV positions. Among existing studies on VLC-enabled UAVs, a graphical approach is utilized to maximize the CTR under covertness, reliability, and flight region constraints [25]. Besides, UAV altitude optimization based on mutual information is explored in [26] and [36]. However, these works rely on fixed relative position assumptions and are not readily extended to dynamic multi-UAV cooperation scenarios.

Motivated by these insights, this work introduces a cooperative jamming UAV to assist the visible light covert communication (VLCC) system, where the 3D trajectory design and resource allocation are jointly optimized to maximize the MACTR. A comprehensive comparison of related UAV-assisted covert communication works is provided in Table I, and our main contributions are as follows.

- An RSMA-based VLCC system model with dual UAVs is developed, where the source UAV transmits confidential data to users, assisted by a jamming UAV that introduces power uncertainty to confuse wardens. Moreover, the detection performance of wardens is analyzed, including both the optimal detection threshold and detection error probability (DEP). Furthermore, the MACTR maximization problem is formulated with various constraints.
- A block coordinate descent (BCD)-based effective algorithm is adopted to tackle the MACTR maximization problem. Specifically, the 3D trajectory design is decomposed into horizontal coordinate arrangement and altitude adjustment subproblems, solved by the successive convex approximation (SCA) method and geometric programming (GP)-based approach, respectively. Meanwhile, the resource allocation, involving power assignment and rate-

TABLE I
WORKS ON UAV-ASSISTED COVERT COMMUNICATIONS

Reference	Band	Role of UAVs	Metric	SDoF ¹	Trajectory Design	MA Scheme	Randomness Source
[5]	RF	BS	Average CTR	2	✓	✗	Environmental Noise
[16]	RF	BS	SNR	3	✗	✗	Gaussian Signaling
[19]	RF	BS	MACTR	2	✓	OMA	Environmental Noise
[20]	RF	BS	MACEE ²	3	✓	OMA	Gaussian Signaling
[21]	RF	BS	Average CTR	3	✓	✗	Gaussian Signaling
[22]	RF	BS	CTR	2	✗	NOMA	Environmental Noise
[23]	RF	BS	MACTR	2	✓	OMA	Gaussian Signaling
[2]	RF	BS and Jammer	MACTR	2	✓	OMA	Gaussian Signaling
[9]	RF	BS and Jammer	Minimum CTR	3	✗	OMA	Jamming Power
[3]	RF	Receiver	Average CTR	2	✓	NOMA	Channel Fading
[7]	RF	Receiver	CTR	2	✗	✗	Channel Fading
[24]	RF	Receiver and Jammer	MACTR	2	✓	OMA	Jamming Power
[8]	RF	Relay	Average CTR	3	✓	✗	Environmental Noise
[17]	RF	Relay	CTR	2	✗	✗	Environmental Noise
[25]	VLC	BS	CTR	2	✗	NOMA	Uniform Signaling
[26]	VLC	BS	Mutual Information	1	✗	✗	Uniform Signaling
Our Work	VLC	BS and Jammer	MACTR	3	✓	RSMA	Jamming Power

¹ SDoF: Spatial degrees of freedom. ² MACEE: Minimum average covert energy efficiency.

splitting, is jointly optimized as one subproblem by the majorization-minimization (MM) algorithm.

- The convergence and computational complexity of the proposed algorithm are analyzed. Extensive simulations demonstrate the significant MACTR enhancement over baseline methods and traditional MA schemes. Besides, key performance metrics and parameters are also investigated, providing insights for UAV-assisted VLCC systems.

The remainder of this paper is organized as follows. Section II introduces the RSMA-based VLCC system with dual UAVs, detailing the system architecture, channel model, and RSMA scheme. In Section III, the DEP and optimal detection threshold for wardens are derived. Then, the MACTR maximization problem and its equivalent transformation are formulated in Section IV. Section V presents the BCD-based algorithm for 3D trajectory design and resource allocation, followed by analyzing the convergence and computational complexity. Moreover, Section VI evaluates the proposed algorithm through simulations, demonstrating its superior performance. Finally, the paper is concluded in Section VII.

Notations: Scalars, vectors, matrices, and sets are denoted by normal letters, boldface lowercase letters, boldface capital letters, and fancy letters, respectively. The symbols $|\cdot|$, $\|\cdot\|$, ∂ , and $(\cdot)^T$ represent the absolute value, 2-norm, partial operator, and transpose, respectively. Besides, $\mathbf{A} \succeq \mathbf{O}$ indicates that each element of matrix $\mathbf{A}$ is non-negative. The operators $\mathbb{E}[\cdot]$ and $\Pr\{\cdot\}$ denote the expectation and probability, respectively.

II. SYSTEM MODEL

In this section, the RSMA-based VLCC system with dual UAVs is presented, including the system architecture, channel model, and RSMA scheme. For clarity, key notations and their definitions are summarized in Table II, and the index $[i]$ indicates that the variable is associated with time slot i .

TABLE II

LIST OF PRIMARY NOTATIONS

Notation	Definition
$k, K, \mathcal{K}$	The index, number, and set of users
$w, W, \mathcal{W}$	The index, number, and set of wardens
T	The flight period
$i, I, \mathcal{I}$	The index, number, and set of time slots
δ	The duration of each time slot
$\mathbf{u}_j$	The horizontal coordinate of ground node j
$\mathbf{q}_v[i]$	The horizontal coordinate of UAV v
$z_v[i]$	The altitude of UAV v
$d_{v,j}[i]$	The slant distance between UAV v and ground node j
$h_{v,j}[i]$	The channel gain between UAV v and ground node j
$P_C[i]$	The transmit power for the common stream
$P_k[i]$	The transmit power for the private stream of user k
$P_S[i]$	The total transmit power of the source UAV S
$P_J[i]$	The transmit power of the jamming UAV J
$R_{C,k}[i]$	The CTR for decoding the common stream at user k
$\underline{R}_{C,k}[i]$	The tight lower bound for the expectation of $R_{C,k}[i]$
$\bar{R}_C[i]$	The achievable common CTR for all users
$C_k[i]$	The portion of the common CTR allocated to user k
$R_k[i]$	The private CTR of user k
$\underline{R}_k[i]$	The tight lower bound for the expectation of $R_k[i]$
$R_{k,T}[i]$	The total CTR of user k
$\bar{R}_k$	The average CTR of user k over all time slots
$\mathcal{H}_0, \mathcal{H}_1$	The null and alternative hypotheses for detection
$Y_w[i]$	The average received power at warden w
$\tau_w[i]$	The detection threshold of warden w
$\xi_w[i]$	The DEP of warden w
$\mathbb{P}_{FA,w}[i]$	The false alarm probability of warden w
$\mathbb{P}_{MD,w}[i]$	The miss detection probability of warden w

A. System Architecture

As illustrated in Fig. 1, the RSMA-based VLCC system consists of dual UAVs and multiple ground nodes, including K users and W non-colluding wardens in sets $\mathcal{K}$ and $\mathcal{W}$, respectively. Each UAV is equipped with an LED for optical signal transmission, while each ground node employs a PD to receive the signal. In particular, source UAV S transmits

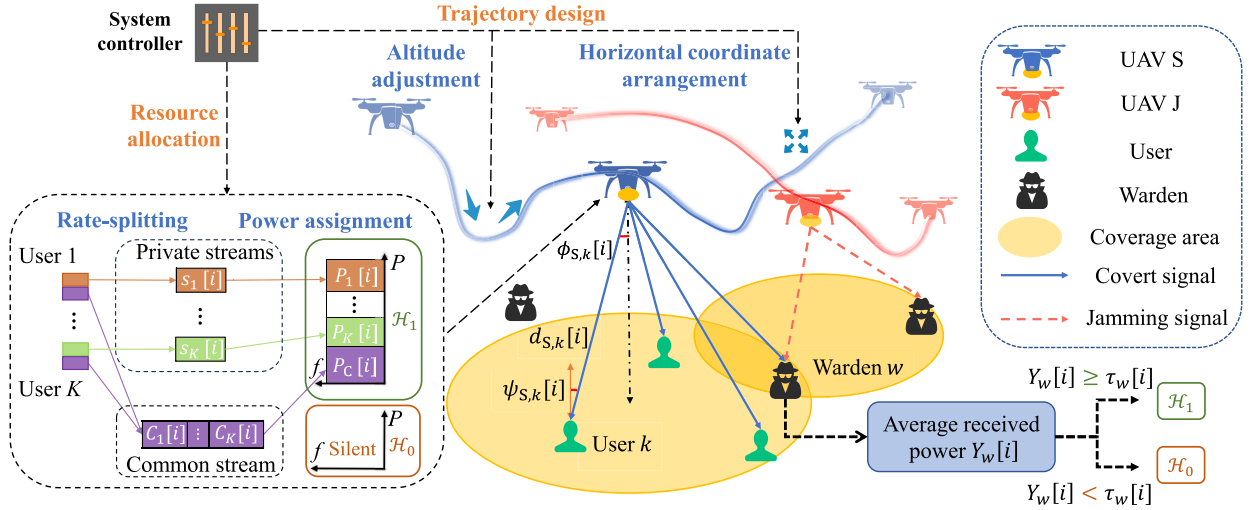


Fig. 1. The architecture of the RSMA-based VLCC with jammer assistance for UAV systems.

confidential data to users while avoiding detection by wardens. To enhance the covertness, an uninformed jammer, i.e., UAV J, simultaneously transmits jamming signals to create ambiguity in the received power of wardens [18].

The 3D Cartesian coordinate system is employed to describe the locations of ground nodes and UAVs. Specifically, the horizontal coordinates of user $k \in \mathcal{K}$ and warden $w \in \mathcal{W}$ are denoted as $\mathbf{u}_k$ and $\mathbf{u}_w$, respectively. For mobile UAVs, the continuous-time trajectory design leads to an infinite number of constraints. To address this, the flight period T is discretized into I equal time slots with duration $\delta = T/I$, where UAV positions remain approximately constant within each time slot. In time slot i , the horizontal coordinate of UAV v is given by $\mathbf{q}_v[i] = (q_{v,1}[i], q_{v,2}[i])^T$, where $v \in \{S, J\}$ and $i \in \{1, 2, \dots, I\} \triangleq \mathcal{I}$. Similarly, $z_v[i]$ represents the altitude of UAV v in time slot i .

B. Channel Model

The high UAV altitudes effectively alleviate the obstacle influence, ensuring reliable LoS links [30]. Additionally, given the significantly higher path loss of non-line-of-sight propagation compared to LoS conditions [33], our analysis exclusively focuses on the dominant LoS component. Therefore, the channel gain between UAV v and ground node $j \in \mathcal{K} \cup \mathcal{W}$ during time slot i is modeled as [35]

$$h_{v,j}[i] = \begin{cases} \frac{(m+1)An_r^2\Upsilon}{2\pi \sin^2 \Psi_c} \frac{\cos^m(\phi_{v,j}[i]) \cos(\psi_{v,j}[i])}{d_{v,j}^2[i]}, & 0 \leq \psi_{v,j}[i] \leq \Psi_c, \\ 0, & \psi_{v,j}[i] > \Psi_c, \end{cases} \quad (1)$$

where m , A , n_r , Υ , and Ψ_c correspond to the Lambert order, detector area, refractive index, optical filter gain, and field-of-view semi-angle, respectively. Meanwhile, $d_{v,j}[i]$, $\phi_{v,j}[i]$, and $\psi_{v,j}[i]$ denote the slant distance, irradiance angle, and incidence angle for the LoS path from UAV v to ground node

j in time slot i , respectively. Besides, the slant distance $d_{v,j}[i]$ is expressed as

$$d_{v,j}[i] = \sqrt{\|\mathbf{q}_v[i] - \mathbf{u}_j\|^2 + z_v^2[i]}. \quad (2)$$

In order to facilitate the theoretical analysis, it is assumed that the normal vectors of LEDs and PDs are perpendicular to the horizontal plane [26], [32], resulting in $\cos \phi_{v,j}[i] = \cos \psi_{v,j}[i] = z_v[i]/d_{v,j}[i]$. Accordingly, when $0 \leq \psi_{v,j}[i] \leq \Psi_c$, the channel gain $h_{v,j}[i]$ can be simplified to

$$h_{v,j}[i] = h_0 z_v^{m+1}[i] d_{v,j}^{-(m+3)}[i], \quad (3)$$

where $h_0 = ((m+1)An_r^2\Upsilon) / (2\pi \sin^2 \Psi_c)$. Additionally, it is assumed that the CSI for legitimate users can be acquired through existing channel estimation techniques [37]. While for wardens, a bounded error CSI model is adopted [8], [38], and its impact is analyzed through simulations in Section VI-B.

Remark: Unlike RF channels, where multipath fading, shadowing, and dynamic interference cause rapid fluctuations, VLC channels in UAV networks are quasi-static and predictable due to LoS-dominant propagation [30], and negligible Doppler shift and small-scale fading [39]. These characteristics make off-line trajectory optimization both effective and practically relevant in the proposed VLCC system.

C. RSMA Scheme

In the RSMA scheme, each user's message is split into a common part and a private part. Subsequently, the common parts of all users are combined into a common message, which is encoded into a unified common stream, denoted by $s_C[i]$ in time slot i . Meanwhile, the private message of each user is encoded into a distinct private stream, represented by $s_k[i]$ for user k . Thus, the transmit signal of UAV S is formulated as

$$x[i] = \sqrt{P_C[i]}s_C[i] + \sum_{k \in \mathcal{K}} \sqrt{P_k[i]}s_k[i], \quad (4)$$

where $P_C[i]$ and $P_k[i]$ denote the transmit power assigned to the common stream $s_C[i]$ and the private stream $s_k[i]$, respectively, with $\mathbb{E}[|s_C[i]|^2] = \mathbb{E}[|s_k[i]|^2] = 1$.

Moreover, UAV J transmits a normalized jamming signal $s_J[i]$ with power $P_J[i]$ in time slot i . Based on prior works in [9] and [18], the jamming power $P_J[i]$ is assumed to follow a uniform distribution over $[0, P_{J,\max}]$, where $P_{J,\max}$ represents the maximum jamming power of UAV J. Therefore, the received signal at user k can be given by

$$y_k[i] = h_{S,k}[i]x[i] + h_{J,k}[i]\sqrt{P_J[i]}s_J[i] + n_k[i], \quad (5)$$

where $n_k[i] \sim \mathcal{N}(0, \sigma_k^2)$ is the zero-mean Gaussian noise at user k with variance σ_k^2 .

According to the RSMA scheme, the common stream is decoded first while treating all private streams as noise [13]. Therefore, the CTR for decoding the common stream at user k in time slot i can be demonstrated as [40]

$$R_{C,k}[i] = \frac{B}{2} \log_2 \left(1 + \frac{\frac{e}{2\pi} h_{S,k}^2[i] P_C[i]}{h_{J,k}^2[i] P_J[i] + h_{S,k}^2[i] \sum_{l \in \mathcal{K}} P_l[i] + \sigma_k^2} \right), \quad (6)$$

where B denotes the bandwidth. However, the randomness of the jamming power $P_J[i]$ makes the analysis of (6) challenging. Therefore, a tight lower bound (LB) for the expectation of (6) is derived as [18]

$$\underline{R}_{C,k}[i] = \frac{B}{2} \log_2 \left(1 + \frac{\frac{e}{2\pi} h_{S,k}^2[i] P_C[i]}{h_{J,k}^2[i] \frac{P_{J,\max}}{2} + h_{S,k}^2[i] \sum_{l \in \mathcal{K}} P_l[i] + \sigma_k^2} \right). \quad (7)$$

Besides, to ensure the common stream $s_C[i]$ is successfully decoded by all users, its data rate can not exceed

$$\underline{R}_C[i] = \min \{ \underline{R}_{C,1}[i], \underline{R}_{C,2}[i], \dots, \underline{R}_{C,K}[i] \}. \quad (8)$$

Since $s_C[i]$ contains sub-messages of all K users, $\underline{R}_C[i]$ is allocated among users proportionally to their sub-messages contributions. Let the common CTR $C_k[i]$ denote the portion of $\underline{R}_C[i]$ assigned to user k in time slot i , which satisfies

$$\sum_{k \in \mathcal{K}} C_k[i] = \underline{R}_C[i]. \quad (9)$$

After decoding the common stream $s_C[i]$, users re-encode and then subtract it from the received signal, which requires successive interference cancellation capabilities and increases the receiver's complexity. Subsequently, the private stream $s_k[i]$ is decoded by user k , treating the remaining signal as noise [13]. Thus, the private CTR of user k is given as

$$R_k[i] = \frac{B}{2} \log_2 \left(1 + \frac{\frac{e}{2\pi} h_{S,k}^2[i] P_k[i]}{h_{J,k}^2[i] P_J[i] + h_{S,k}^2[i] \sum_{l \in \mathcal{K}_k} P_l[i] + \sigma_k^2} \right), \quad (10)$$

where $\mathcal{K}_k = \mathcal{K} / \{k\}$ is the set of all users excluding user k . Similarly, a tight LB for the expectation of (10) can be expressed as

$$\underline{R}_k[i] = \frac{B}{2} \log_2 \left(1 + \frac{\frac{e}{2\pi} h_{S,k}^2[i] P_k[i]}{h_{J,k}^2[i] \frac{P_{J,\max}}{2} + h_{S,k}^2[i] \sum_{l \in \mathcal{K}_k} P_l[i] + \sigma_k^2} \right). \quad (11)$$

Following that, the total CTR of user k in time slot i consists of both common and private CTRs, which is given by

$$R_{k,T}[i] = C_k[i] + \underline{R}_k[i]. \quad (12)$$

Then, the average CTR of user k among all time slots and the CTR of UAV S in time slot i are defined as

$$\bar{R}_k = \frac{1}{I} \sum_{i=1}^I R_{k,T}[i], \quad (13)$$

$$R_S[i] = \sum_{k \in \mathcal{K}} R_{k,T}[i]. \quad (14)$$

III. COVERTNESS ANALYSIS

In this section, the binary hypothesis testing of wardens is discussed in Section III-A. Then, the detection performance of wardens is analyzed in Section III-B, including both the optimal detection threshold and the corresponding DEP.

A. Binary Hypothesis Testing

Considering the worst-case scenario for covert communications, it is assumed that wardens possess the complete statistical knowledge, including the 3D coordinates of UAVs, the distribution of jamming power, and the noise variance [5], [18], [41].

Each warden conducts a binary hypothesis test independently to determine whether UAV S is transmitting confidential information to users. The null hypothesis $\mathcal{H}_0$ indicates that UAV S is silent, while the alternative hypothesis $\mathcal{H}_1$ signifies that UAV S is engaged in covert communications. Under hypotheses $\mathcal{H}_0$ and $\mathcal{H}_1$, the received signal at warden w in time slot i is given by

$$y_w[i] = \begin{cases} h_{J,w}[i] \sqrt{P_J[i]} s_J[i] + n_w[i], & \mathcal{H}_0, \\ h_{S,w}[i] x[i] + h_{J,w}[i] \sqrt{P_J[i]} s_J[i] + n_w[i], & \mathcal{H}_1, \end{cases} \quad (15)$$

where $n_w[i] \sim \mathcal{N}(0, \sigma_w^2)$ is the zero-mean Gaussian noise with variance σ_w^2 .

As mentioned in [22] and [25], the number of channel uses is assumed to approach infinity. This assumption establishes a theoretical upper bound on the number of samples that a warden can receive in practice and is well-justified for two primary reasons. First, the number of symbols transmitted per time slot is proportional to the communication bandwidth [17], which is abundant in optical band. Second, a significant timescale disparity exists between the motion of UAVs and symbol transmission, allowing for a sufficiently large number of samples in each time slot [24].

Consequently, by applying the strong law of large numbers, the average power received at warden w is expressed as

$$Y_w[i] = |y_w[i]|^2 = \begin{cases} P_J[i]h_{J,w}^2[i] + \sigma_w^2, & \mathcal{H}_0, \\ P_S[i]h_{S,w}^2[i] + P_J[i]h_{J,w}^2[i] + \sigma_w^2, & \mathcal{H}_1, \end{cases} \quad (16)$$

where $P_S[i] \triangleq P_C[i] + \sum_{k \in \mathcal{K}} P_k[i]$ represents the total transmit power of UAV S in time slot i .

Thereafter, the decision rule for warden w that minimizes the DEP is given by

$$Y_w[i] \underset{\mathcal{D}_0}{\overset{\mathcal{D}_1}{\gtrless}} \tau_w[i], \quad (17)$$

where τ_w represents the detection threshold, while $\mathcal{D}_0$ and $\mathcal{D}_1$ are the detection results in favor of $\mathcal{H}_0$ and $\mathcal{H}_1$, respectively.

B. Detection Performance

Based on the above discussions, the false alarm probability and the miss detection probability can be expressed as $\mathbb{P}_{FA,w}[i] = \Pr\{Y_w[i] > \tau_w[i] \mid \mathcal{H}_0\}$ and $\mathbb{P}_{MD,w}[i] = \Pr\{Y_w[i] \leq \tau_w[i] \mid \mathcal{H}_1\}$, respectively. Consequently, the DEP can be written as

$$\xi_w[i] = \mathbb{P}_{FA,w}[i] + \mathbb{P}_{MD,w}[i]. \quad (18)$$

Substituting (16) into $\mathbb{P}_{FA,w}[i]$, the false alarm probability is demonstrated as

$$\begin{aligned} \mathbb{P}_{FA,w}[i] &= \Pr\left\{P_J[i] > \frac{\tau_w[i] - \sigma_w^2}{h_{J,w}^2[i]}\right\} \\ &= \begin{cases} 1, & \tau_w[i] < \sigma_w^2, \\ 1 - \frac{\tau_w[i] - \sigma_w^2}{\Xi_{w,1}[i]}, & \sigma_w^2 \leq \tau_w[i] \leq \Xi_{w,1}[i] + \sigma_w^2, \\ 0, & \tau_w[i] > \Xi_{w,1}[i] + \sigma_w^2, \end{cases} \end{aligned} \quad (19)$$

where $\Xi_{w,1}[i] = h_{J,w}^2[i]P_{J,\max}$. Likewise, the miss detection probability is represented by

$$\begin{aligned} \mathbb{P}_{MD,w}[i] &= \Pr\left\{P_J[i] \leq \frac{\tau_w[i] - \sigma_w^2 - h_{S,w}^2[i]P_S[i]}{h_{J,w}^2[i]}\right\} \\ &= \begin{cases} 0, & \tau_w[i] < \Xi_{w,2}[i] + \sigma_w^2, \\ \frac{\tau_w[i] - \sigma_w^2 - \Xi_{w,2}[i]}{\Xi_{w,1}[i]}, & \Xi_{w,2}[i] + \sigma_w^2 \leq \tau_w[i] \leq \Xi_{w,1}[i] + \Xi_{w,2}[i] + \sigma_w^2, \\ 1, & \tau_w[i] > \Xi_{w,1}[i] + \Xi_{w,2}[i] + \sigma_w^2, \end{cases} \end{aligned} \quad (20)$$

where $\Xi_{w,2}[i] = h_{S,w}^2[i]P_S[i]$. In the following, the optimal detection threshold and the minimum DEP are presented in Theorem 1.

Theorem 1: The optimal detection threshold $\tau_w^*[i]$ and the corresponding DEP $\xi_w^*[i]$ of warden w are given by

$$\begin{aligned} \tau_w^*[i] &\in [\min\{\Xi_{w,1}[i], \Xi_{w,2}[i]\} + \sigma_w^2, \\ &\quad \max\{\Xi_{w,1}[i], \Xi_{w,2}[i]\} + \sigma_w^2], \end{aligned} \quad (21)$$

$$\xi_w^*[i] = \begin{cases} 0, & \Xi_{w,1}[i] < \Xi_{w,2}[i], \\ 1 - \frac{\Xi_{w,2}[i]}{\Xi_{w,1}[i]}, & \Xi_{w,1}[i] \geq \Xi_{w,2}[i]. \end{cases} \quad (22)$$

Proof: The detailed proof is provided in Appendix A. ■

It is important to note that if $\Xi_{w,1}[i] < \Xi_{w,2}[i]$, warden w can achieve a zero DEP. Therefore, we only focus on the non-trivial case where $\Xi_{w,1}[i] \geq \Xi_{w,2}[i]$.

IV. PROBLEM FORMULATION

For clarity, the horizontal coordinate matrix and altitude vector of UAV v are denoted by $\mathbf{Q}_v = \{\mathbf{q}_v[i] \mid i \in \mathcal{I}\}$ and $\mathbf{z}_v = \{z_v[i] \mid i \in \mathcal{I}\}$, respectively. Additionally, the transmit power matrix of UAV S is given by $\mathbf{P} = \{P_C[i]; P_k[i] \mid i \in \mathcal{I}, k \in \mathcal{K}\}$, while the common CTR matrix of users is defined as $\mathbf{C} = \{C_k[i] \mid i \in \mathcal{I}, k \in \mathcal{K}\}$. Therefore, by joint trajectory design and resource allocation, the MACTR maximization problem for the RSMA-based VLCC in the UAV system is summarized as

$$(\mathbf{P0}) : \max_{\mathbf{Q}_v, \mathbf{z}_v, \mathbf{P}, \mathbf{C}} \min_k \bar{R}_k \quad (23)$$

$$\text{s.t. } \mathbf{q}_v[1] = \mathbf{q}_{v,I}, \mathbf{q}_v[I] = \mathbf{q}_{v,F}, \quad (23a)$$

$$\|\mathbf{q}_v[i+1] - \mathbf{q}_v[i]\| \leq V_{H,\max}\delta, \quad \forall i \in \mathcal{I}/\{I\}, \quad (23b)$$

$$\|\mathbf{q}_S[i] - \mathbf{q}_J[i]\| \geq r_{\min}, \quad \forall i \in \mathcal{I}, \quad (23c)$$

$$z_v[1] = z_{v,I}, z_v[I] = z_{v,F}, \quad (23d)$$

$$|z_v[i+1] - z_v[i]| \leq V_{A,\max}\delta, \quad \forall i \in \mathcal{I}/\{I\}, \quad (23e)$$

$$z_{\min} \leq z_v[i] \leq z_{\max}, \quad \forall i \in \mathcal{I}, \quad (23f)$$

$$\mathbf{P} \succeq \mathbf{O}, \quad (23g)$$

$$P_S[i] \leq P_{S,\max}, \quad \forall i \in \mathcal{I}, \quad (23h)$$

$$\frac{1}{I} \sum_{i=1}^I P_S[i] \leq \bar{P}_{\max}, \quad (23i)$$

$$\mathbf{C} \succeq \mathbf{O}, \quad (23j)$$

$$\sum_{k \in \mathcal{K}} C_k[i] = \underline{R}_C[i], \quad \forall i \in \mathcal{I}, \quad (23k)$$

$$\xi_w^*[i] \geq 1 - \epsilon, \quad \forall w \in \mathcal{W}, i \in \mathcal{I}. \quad (23l)$$

The constraints are explained as follows.

- 1) **Horizontal coordinate constraints:** Constraint (23a) anchors the UAV flight paths to the operational scope of a given mission by defining the initial position $\mathbf{q}_{v,I}$ and final position $\mathbf{q}_{v,F}$ of UAV v . Meanwhile, constraint (23b) reflects the maximum horizontal speed of $V_{H,\max}$, ensuring the designed trajectory is achievable by the UAV propulsion system. Finally, to guarantee safe co-deployment in the dual-UAV system, constraint (23c) imposes a minimum separation distance $r_{\min}$, which is crucial for collision avoidance.
- 2) **Altitude constraints:** Constraints (23d) and (23e) specify the initial altitude $z_{v,I}$, final altitude $z_{v,F}$, and maximum vertical speed $V_{A,\max}$ restricted by the mechanical capabilities. Besides, $z_{\min}$ and $z_{\max}$ define the minimum and maximum allowable altitudes in (23f),

respectively, to ensure LoS links and sufficient channel gain.

- 3) **Power assignment constraints:** Constraint (23g) ensures that the power assigned to common and private streams is non-negative, reflecting the physical property of optical signals. Moreover, the peak power of UAV S cannot surpass the maximum output of the LED transmitter $P_{S,\max}$, as demonstrated in (23h). The average power constraint (23i) governs the long-term power budget $\bar{P}_{\max}$ for communication-related tasks of UAV S.
- 4) **RSMA decoding constraints:** Constraints (23j) and (23k) guarantee successful decoding of the common stream through non-negative rate-splitting.
- 5) **Coverttness constraint:** Constraint (23l) specifies the coverttness threshold $\epsilon \in (0, 1]$, which translates the requirement of covert communication into a tangible performance metric.

Remark: In the formulated problem (P0), the motion of UAVs is decomposed into horizontal and vertical directions. Its physical feasibility is grounded in the standard hierarchical control architecture of UAVs [42]. In this framework, the high-level optimization algorithm is responsible for generating the trajectory, while a dedicated low-level onboard controller is assumed to handle the complex flight dynamics and coupling effects to accurately track the planned trajectory, utilizing advanced control techniques such as those in [43] and [44]. Besides, this strategy makes the optimization problem computationally tractable.

Due to the difficulty of solving problem (P0) in the max-min form, auxiliary variables χ and $\varsigma = \{\varsigma[i] \mid i \in \mathcal{I}\}$ are introduced to transform problem (P0) as

$$(\mathbf{P0-1}) : \max_{\mathbf{Q}_v, \mathbf{z}_v, \mathbf{P}, \mathbf{C}, \chi, \varsigma} \chi \quad (24)$$

$$\text{s.t. } \bar{R}_k \geq \chi, \quad \forall k \in \mathcal{K}, \quad (24a)$$

$$\underline{R}_{C,k}[i] \geq \varsigma[i], \quad \forall k \in \mathcal{K}, i \in \mathcal{I}, \quad (24b)$$

$$\sum_{k \in \mathcal{K}} C_k[i] = \varsigma[i], \quad \forall i \in \mathcal{I}, \quad (24c)$$

$$P_S[i] \frac{h_{S,w}^2[i]}{h_{J,w}^2[i]} \leq P_{J,\max} \epsilon, \quad \forall w \in \mathcal{W}, i \in \mathcal{I}, \quad (24d)$$

$$(23a) \sim (23j).$$

The equivalence of these two problems will be discussed in the following theorem.

Theorem 2: Problem (P0-1) is equivalent to the problem (P0).

Proof: Problem (P0-1) differs from (P0) in objective function (24) and constraints (24a)~(24d). Their equivalence is established through the following analysis. For (24) and (24a), the optimal solution of problem (P0-1) must satisfy $\chi = \min_k \bar{R}_k$. Otherwise, $\min_k \bar{R}_k$ would provide a better solution, leading to a contradiction. Moreover, constraints (24b) and (24c) ensure $\sum_{k \in \mathcal{K}} C_k[i] = \varsigma[i] \leq \underline{R}_C[i]$, which is a relaxed form compared to the original constraint (23k), due to the inequality. While this relaxation might theoretically

permit higher χ , the upper bound of χ remains constrained by the invariant upper bound of $C_k[i]$. Besides, constraint (24d) can be derived from (23l) through basic algebraic operations. Therefore, the optimal χ does not change, which completes the proof. ■

V. PROPOSED MACTR MAXIMIZATION ALGORITHM

The transformed problem (P0-1) remains challenging due to the non-convex constraints (23c), (24a), (24b), and (24d). To address this, a BCD-based algorithm is proposed to decompose the problem (P0-1) into multiple subproblems. Specifically, the 3D trajectory design is decomposed into horizontal coordinate arrangement and altitude adjustment subproblems, solved by the SCA method and GP-based approach in Sections V-A and V-B, respectively. Meanwhile, the resource allocation, involving power assignment and rate-splitting, is jointly optimized by the MM algorithm in Section V-C. Besides, the convergence and computational complexity of the proposed algorithm are analyzed in Sections V-D and V-E, respectively.

Although the BCD framework is adopted, each subproblem remains non-convex. Specifically, the trajectory design is particularly challenging due to non-convex polynomial terms in the SINR expression. To address this, auxiliary variables are first introduced to decompose the complex constraints into simpler ones, whose convexity is analyzed before applying approximation techniques for convexification. Besides, in the altitude adjustment subproblem, a variable substitution is also applied to handle the monomial-like structure. In contrast, the resource allocation subproblem features more manageable linear-fractional structures, for which surrogate functions and related approximations are applied directly. When alternately solving these subproblems, the coupling arises as the UAVs' 3D trajectories determine the channel gains for resource allocation, which in turn defines the objective function that guides subsequent trajectory design. The detailed analyses are provided in the following.

A. Horizontal Coordinate Arrangement

Focusing on the horizontal coordinate arrangement, problem (P0-1) with given $\mathbf{z}_v$, $\mathbf{P}$, $\mathbf{C}$, and ς is transformed into

$$(\mathbf{P1}) : \max_{\mathbf{Q}_v, \chi} \chi \quad (25)$$

$$\text{s.t. } (23a) \sim (23c), (24a), (24b), (24d).$$

The constraints (24a), (24b), and (24d) involve coupled fractional terms that are nonlinear functions of $\mathbf{Q}_v$, which are challenging to optimize directly. To address this, auxiliary variables are introduced to decouple the terms and transform them into an expanded set of more manageable constraints, as stated in Theorem 3.

Theorem 3: By introducing the auxiliary sets $\mathfrak{N}_H \triangleq \{\zeta_{k,H}[i], \vartheta_{k,H}[i], t_{k,H}[i] \mid k \in \mathcal{K}, i \in \mathcal{I}\}$ and $\mathfrak{J}_H \triangleq \{\iota_{w,H}[i], \kappa_{w,H}[i] \mid w \in \mathcal{W}, i \in \mathcal{I}\}$, constraints (24a), (24b), and (24d) can be equivalently transformed into

$$\frac{1}{I} \sum_{i=1}^I \left(\frac{B}{2} \log_2 \left(1 + \frac{\frac{e}{2\pi} P_k[i]}{\zeta_{k,H}^{-1}[i] + \sum_{l \in \mathcal{K}_k} P_l[i]} \right) + C_k[i] \right) \geq \chi, \quad (26)$$

$$\frac{B}{2} \log_2 \left(1 + \frac{\frac{e}{2\pi} P_C[i]}{\zeta_{k,H}^{-1}[i] + \sum_{l \in \mathcal{K}} P_l[i]} \right) \geq \varsigma[i], \quad (27)$$

$$\frac{\vartheta_{k,H}^2[i]}{\zeta_{k,H}^2[i]} \geq t_{k,H}[i] + \sigma_k^2, \quad (28)$$

$$\frac{1}{\vartheta_{k,H}[i]} \geq \frac{d_{S,k}^{m+3}[i]}{D_k[i]}, \quad (29)$$

$$\left(\frac{E_k[i]}{t_{k,H}[i]} \right)^{\frac{1}{m+3}} \leq d_{J,k}^2[i], \quad (30)$$

$$\frac{\iota_{w,H}^2[i]}{\kappa_{w,H}[i]} \leq \sqrt{\frac{P_{J,\max}}{P_S[i]}} \epsilon, \quad (31)$$

$$\left(\frac{h_0 z_S^{m+1}[i]}{\iota_{w,H}^2[i]} \right)^{\frac{2}{m+3}} \leq d_{S,w}^2[i], \quad (32)$$

$$\frac{h_0 z_J^{m+1}[i]}{\kappa_{w,H}[i]} \geq d_{J,w}^{m+3}[i], \quad (33)$$

where $D_k[i] = h_0 z_S^{m+1}[i]$ and $E_k[i] = \frac{1}{2} P_{J,\max} h_0^2 z_J^{2m+2}[i]$ are constants.

Proof: The detailed proof is provided in Appendix B. ■

Then, the convexity properties of (26)~(33) will be discussed in the following lemma.

Lemma 1: Constraints (26), (27) and (31) are convex.

Proof: The detailed proof is provided in Appendix C. ■

Building on Lemma 1, constraints (26), (27) and (31) are convex, whereas constraints (23c), (28)~(30), (32) and (33) still exhibit non-convexity. Therefore, the SCA method is employed, which iteratively constructs subproblems by replacing non-convex feasible sets with inner convex approximations [45]. To this end, first-order Taylor expansions are applied to the convex components in (23c), (28)~(30), (32), and (33), which are listed as

$$2(\hat{\mathbf{q}}_S[i] - \hat{\mathbf{q}}_J[i])^T (\mathbf{q}_S[i] - \mathbf{q}_J[i]) - \|\hat{\mathbf{q}}_S[i] - \hat{\mathbf{q}}_J[i]\|^2 \geq r_{\min}^2, \quad (34)$$

$$\frac{2\hat{\vartheta}_{k,H}[i]}{\hat{\zeta}_{k,H}[i]} \vartheta_{k,H}[i] - \frac{\hat{\vartheta}_{k,H}^2[i]}{\hat{\zeta}_{k,H}^2[i]} \zeta_{k,H}[i] \geq t_{k,H}[i] + \sigma_k^2, \quad (35)$$

$$-\frac{\vartheta_{k,H}[i]}{\hat{\vartheta}_{k,H}^2[i]} + \frac{2}{\hat{\vartheta}_{k,H}[i]} \geq \frac{d_{S,k}^{m+3}[i]}{D_k[i]}, \quad (36)$$

$$\left(\frac{E_k[i]}{t_{k,H}[i]} \right)^{\frac{1}{m+3}} \leq z_J^2[i] + F_i(\mathbf{Q}_J; \hat{\mathbf{Q}}_J, \mathbf{u}_k), \quad (37)$$

$$\left(\frac{h_0 z_S^{m+1}[i]}{\iota_{w,H}^2[i]} \right)^{\frac{2}{m+3}} \leq z_S^2[i] + F_i(\mathbf{Q}_S; \hat{\mathbf{Q}}_S, \mathbf{u}_w), \quad (38)$$

$$h_0 z_J^{m+1}[i] \left(-\frac{\kappa_{w,H}[i]}{\hat{\kappa}_{w,H}^2[i]} + \frac{2}{\hat{\kappa}_{w,H}[i]} \right) \geq d_{J,w}^{m+3}[i], \quad (39)$$

where $\hat{\mathbf{Q}}_S$, $\hat{\mathbf{Q}}_J$, $\hat{\vartheta}_{k,H}[i]$, $\hat{\zeta}_{k,H}[i]$ and $\hat{\kappa}_{w,H}[i]$ are obtained from the last iteration, and the auxiliary function F_i is defined as

$$F_i(\mathbf{Q}_v; \hat{\mathbf{Q}}_v, \mathbf{u}_j) \triangleq (\hat{\mathbf{q}}_v[i] - \mathbf{u}_j)^T (2q_v[i] - \hat{\mathbf{q}}_v[i] - \mathbf{u}_j). \quad (40)$$

In summary, the problem (P1) can be replaced by

$$\begin{aligned} \text{(P1-1)} : \quad & \max_{\mathbf{Q}_v, \chi, \mathbb{N}_H, \mathbb{J}_H} \chi \\ \text{s.t.} \quad & (23a), (23b), (26), (27), (31), (34) \sim (39), \end{aligned} \quad (41)$$

where $\mathbb{N}_H \triangleq \{\zeta_{k,H}[i], \vartheta_{k,H}[i], t_{k,H}[i] \mid k \in \mathcal{K}, i \in \mathcal{I}\}$ and $\mathbb{J}_H \triangleq \{\iota_{w,H}[i], \kappa_{w,H}[i] \mid w \in \mathcal{W}, i \in \mathcal{I}\}$. As all the constraints are transformed into convex forms, the problem (P1-1) is a convex optimization problem and can be efficiently solved by the interior point method (IPM) [46].

B. Altitude Adjustment

For fixed $\mathbf{Q}_v$, $\mathbf{P}$, $\mathbf{C}$, and ς , the altitude adjustment subproblem can be expressed as

$$\begin{aligned} \text{(P2)} : \quad & \max_{\mathbf{z}_v, \chi} \chi \\ \text{s.t.} \quad & (23d) \sim (23f), (24a), (24b), (24d), \end{aligned} \quad (42)$$

where non-convex constraints (24a), (24b) and (24d) will be addressed subsequently.

The reformulation of (24a), (24b), and (24d) in the altitude adjustment presents additional challenges compared to the horizontal case. In particular, the channel gain expression in (3) is coupled with both the UAV altitude and the slant distance, where the latter is also a function of altitude. This intricate coupling prevents the direct application of the methodology in Section V-A. Furthermore, the channel gain exhibits monomial-like structures [47], which naturally motivates a reformulation as a GP problem in Theorem 4.

Theorem 4: By introducing auxiliary variable sets $\mathbb{N}_A \triangleq \{\zeta_{k,A}[i], \vartheta_{k,A}[i], t_{k,A}[i] \mid k \in \mathcal{K}, i \in \mathcal{I}\}$, $\mathbb{J}_A \triangleq \{\iota_{w,A}[i], \kappa_{w,A}[i] \mid w \in \mathcal{W}, i \in \mathcal{I}\}$ and applying variable substitution $z_v[i] = e^{-\kappa_v[i]}$, constraints (24a), (24b), and (24d) can be equivalently transformed into the GP-compatible form as

$$\frac{1}{I} \sum_{i=1}^I \left(\frac{B}{2} \log_2 \left(1 + \frac{\frac{e}{2\pi} P_k[i]}{\zeta_{k,A}^{-1}[i] + \sum_{l \in \mathcal{K}_k} P_l[i]} \right) + C_k[i] \right) \geq \chi, \quad (43)$$

$$\frac{B}{2} \log_2 \left(1 + \frac{\frac{e}{2\pi} P_C[i]}{\zeta_{k,A}^{-1}[i] + \sum_{l \in \mathcal{K}} P_l[i]} \right) \geq \varsigma[i], \quad (44)$$

$$\frac{\vartheta_{k,A}^2[i]}{\zeta_{k,A}^2[i]} \geq t_{k,A}[i] + \sigma_k^2, \quad (45)$$

$$-2 \ln \vartheta_{k,A}[i] \geq G_i(\kappa_S; \mathbf{Q}_S, \mathbf{u}_k), \quad (46)$$

$$\ln t_{k,A}[i] \geq -G_i(\kappa_J; \mathbf{Q}_J, \mathbf{u}_k) + \ln P_{J,\max} - \ln 2, \quad (47)$$

$$\frac{\iota_{w,A}^2[i]}{\kappa_{w,A}[i]} \leq \sqrt{\frac{P_{J,\max}}{P_S[i]}} \epsilon, \quad (48)$$

$$4 \ln \iota_{w,A}[i] \geq -G_i(\kappa_S; \mathbf{Q}_S, \mathbf{u}_w), \quad (49)$$

$$2 \ln \kappa_{w,A}[i] \leq -G_i(\kappa_J; \mathbf{Q}_J, \mathbf{u}_w), \quad (50)$$

where

$$\begin{aligned} G_i(\kappa_v; \mathbf{Q}_v, \mathbf{u}_j) & \triangleq -\ln h_{v,j}^2[i] = 2(m+1)\kappa_v[i] - 2 \ln h_0 \\ & + (m+3) \ln (\|\mathbf{q}_v[i] - \mathbf{u}_j\|^2 + e^{-2\kappa_v[i]}) \end{aligned} \quad (51)$$

with $\kappa_v = \{\kappa_v[i] \mid i \in \mathcal{I}\}$.

Proof: The detailed proof is provided in Appendix D. ■

Moreover, as original variable z_v is substituted by κ_v , altitude constraints (23d)~(23f) are rewritten as

$$\kappa_v[1] = -\ln z_{v,I}, \quad \kappa_v[I] = -\ln z_{v,F}, \quad (52)$$

$$-V_{A,\max}\delta \leq e^{-\kappa_v[i+1]} - e^{-\kappa_v[i]} \leq V_{A,\max}\delta, \quad (53)$$

$$e^{\kappa_v[i]} z_{\min} \leq 1, \quad e^{-\kappa_v[i]} \leq z_{\max}. \quad (54)$$

Having demonstrated the equivalence of the constraints, we proceed to analyze their convexity properties.

Lemma 2: $G_i(\kappa_v; \mathbf{Q}_v, \mathbf{u}_j)$ is a convex function.

Proof: The second derivative of G_i is calculated as

$$\frac{\partial^2 G_i}{\partial \kappa_v^2[i]} = \frac{4(m+3) \|\mathbf{q}_v[i] - \mathbf{u}_j\|^2 e^{2\kappa_v[i]}}{\left(e^{2\kappa_v[i]} \|\mathbf{q}_v[i] - \mathbf{u}_j\|^2 + 1\right)^2} \geq 0, \quad (55)$$

establishing the convexity of G_i . ■

Building on Lemma 1 and Lemma 2, constraints (43), (44), (48), (52), and (54) are convex, whereas (45)~(47), (49), (50), and (53) still exhibit non-convexity, requiring approximations to enable efficient optimization. Therefore, the LB of G_i is first obtained from $\hat{\kappa}_v$ in the previous iteration, yielding $\underline{G}_i(\kappa_v; \hat{\kappa}_v, \mathbf{Q}_v, \mathbf{u}_j)$ in (56), as shown at the bottom of the page. Then, the aforementioned non-convex constraints are converted to inequalities as

$$\frac{2\hat{\vartheta}_{k,A}[i]}{\hat{\zeta}_{k,A}[i]} \vartheta_{k,A}[i] - \frac{\hat{\vartheta}_{k,A}^2[i]}{\hat{\zeta}_{k,A}^2[i]} \zeta_{k,A}[i] \geq t_{k,A}[i] + \sigma_k^2, \quad (57)$$

$$2 \frac{\hat{\vartheta}_{k,A}[i] - \vartheta_{k,A}[i]}{\hat{\vartheta}_{k,A}[i]} - 2 \ln \hat{\vartheta}_{k,A}[i] \geq G_i(\kappa_S; \mathbf{Q}_S, \mathbf{u}_k), \quad (58)$$

$$\ln t_{k,A}[i] \geq -\underline{G}_i(\kappa_J; \hat{\kappa}_J, \mathbf{Q}_J, \mathbf{u}_k) + \ln P_{J,\max} - \ln 2, \quad (59)$$

$$-V_{A,\max}\delta \leq e^{-\hat{\kappa}_v[i+1]}(1 - \kappa_v[i+1] + \hat{\kappa}_v[i+1]) - e^{-\kappa_v[i]}, \quad (60a)$$

$$e^{-\kappa_v[i+1]} - e^{-\hat{\kappa}_v[i]}(1 - \kappa_v[i] + \hat{\kappa}_v[i]) \leq V_{A,\max}\delta, \quad (60b)$$

$$4 \ln \iota_{w,A}[i] \geq -\underline{G}_i(\kappa_S; \hat{\kappa}_S, \mathbf{Q}_S, \mathbf{u}_w), \quad (61)$$

$$2 \ln \hat{\kappa}_{w,A}[i] + 2 \frac{\kappa_{w,A}[i] - \hat{\kappa}_{w,A}[i]}{\hat{\kappa}_{w,A}[i]} \leq -G_i(\kappa_J; \mathbf{Q}_J, \mathbf{u}_w), \quad (62)$$

where $\hat{\vartheta}_{k,A}[i]$, $\hat{\zeta}_{k,A}[i]$ and $\hat{\kappa}_{w,A}[i]$ are obtained from the last iteration. Moreover, the convexity of above constraints (57)~(62) directly follows from Lemma 1 and Lemma 2.

Therefore, the transformed altitude adjustment problem constructed by the GP-based approach is demonstrated as

$$\begin{aligned} (\mathbf{P2-1}) : \quad & \max_{\kappa_v, \chi, \mathbf{N}_A, \mathbf{J}_A} \chi \\ \text{s.t.} \quad & (43), (44), (48), (52), (54), (57) \sim (62). \end{aligned} \quad (63)$$

Based on the preceding analysis, the problem (P2-1) is solvable by convex optimization algorithms, yielding κ_v^* . Then, the corresponding UAV altitude is recovered as $z_v^* = e^{-\kappa_v^*}$.

C. Power Assignment and Rate-Splitting

Assuming $\mathbf{Q}_v$ and z_v are fixed, the subproblem for power assignment and rate-splitting is formulated as

$$\begin{aligned} (\mathbf{P3}) : \quad & \max_{\mathbf{P}, \mathbf{C}, \chi, \varsigma} \chi \\ \text{s.t.} \quad & (23g) \sim (23j), (24a) \sim (24d). \end{aligned} \quad (64)$$

As constraints (24a) and (24b) are the only non-convex components, the MM algorithm is applied by constructing an approximated subproblem based on the global LBs of (24a) and (24b). Unlike SCA, which allows local approximations, MM requires a global LB to guarantee convergence to a Karush-Kuhn-Tucker point through iterative optimization [48]. To find proper global LBs, the following lemma is introduced.

Lemma 3: For any non-negative ρ and $\hat{\rho}$, the inequality $\log_2(1 + \rho) \geq \alpha \log_2(\rho) + \beta$ holds, where the coefficients α and β are defined as

$$\alpha = \frac{\hat{\rho}}{1 + \hat{\rho}}, \quad (65)$$

$$\beta = \log_2(1 + \hat{\rho}) - \alpha \log_2 \hat{\rho}. \quad (66)$$

Moreover, the function values and first-order derivatives on both sides of the inequality are identical at $\rho = \hat{\rho}$ [31].

Constraint (24a) is first addressed by applying the LB from Lemma 3 to its LHS, yielding

$$\begin{aligned} \frac{1}{I} \sum_{i=1}^I \left(\frac{B}{2} \left(\alpha_k[i] \left(\log_2 \left(\frac{e}{2\pi} h_{S,k}^2[i] P_k[i] \right) \right. \right. \right. \\ \left. \left. \left. - \log_2(\Omega_i(\mathbf{P})) \right) \right) + \beta_k[i] \right) + C_k[i] \geq \chi, \end{aligned} \quad (67)$$

$$\Omega_i(\mathbf{P}) \triangleq h_{J,k}^2[i] \frac{P_{J,\max}}{2} + h_{S,k}^2[i] \sum_{l \in \mathcal{K}_k} P_l[i] + \sigma_k^2. \quad (68)$$

The coefficients $\alpha_k[i]$ and $\beta_k[i]$ are calculated according to (65) and (66) with

$$\hat{\rho}_k[i] = \frac{\frac{e}{2\pi} h_{S,k}^2[i] \hat{P}_k[i]}{\Omega_i(\hat{\mathbf{P}})}, \quad (69)$$

where $\hat{P}_k[i]$ and $\hat{\mathbf{P}}$ are from the solution of the last iteration. As constraint (67) is non-convex, a first-order Taylor expansion of $\log_2(\Omega_i(\mathbf{P}))$ at $\hat{\mathbf{P}}$ is utilized. Due to the concavity of $\log_2(\Omega_i(\mathbf{P}))$, this expansion provides an upper bound for (67) displayed as (70), as shown at the bottom of the next page.

Applying analogous techniques, constraint (24b) can also be transformed into (71), as shown at the bottom of the next page, where the auxiliary function $\Omega_{C,i}(\mathbf{P})$ is defined as

$$\Omega_{C,i}(\mathbf{P}) \triangleq h_{J,k}^2[i] \frac{P_{J,\max}}{2} + h_{S,k}^2[i] \sum_{l \in \mathcal{K}} P_l[i] + \sigma_k^2. \quad (72)$$

Besides, the coefficients $\alpha_{C,k}[i]$ and $\beta_{C,k}[i]$ in (71) are calculated according to (65) and (66) with

$$\hat{\rho}_{C,k}[i] = \frac{e}{2\pi} \frac{h_{S,k}^2[i] \hat{P}_C[i]}{\Omega_{C,i}(\hat{\mathbf{P}})}. \quad (73)$$

$$\underline{G}_i(\kappa_v; \hat{\kappa}_v, \mathbf{Q}_v, \mathbf{u}_j) = 2(m+1)\kappa_v[i] - 2 \ln h_0 + (m+3) \left(\ln \left(\|\mathbf{q}_v[i] - \mathbf{u}_j\|^2 + e^{-2\hat{\kappa}_v[i]} \right) - \frac{2e^{-2\hat{\kappa}_v[i]} (\kappa_v[i] - \hat{\kappa}_v[i])}{\|\mathbf{q}_v[i] - \mathbf{u}_j\|^2 + e^{-2\hat{\kappa}_v[i]}} \right). \quad (56)$$

Accordingly, the problem **(P3)** is approximated as

$$\begin{aligned} \textbf{(P3-1)} : \quad & \max_{\mathbf{P}, \mathbf{C}, \chi, \varsigma} \chi \\ \text{s.t.} \quad & (23g) \sim (23j), (24c), (24d), (70), (71). \end{aligned} \quad (74)$$

Since constraints (23g)~(23j), (24c) and (24d) are affine, while the LHSs of (70) and (71) consist of the sum of a logarithmic function and an affine function, all constraints of problem **(P3-1)** are confirmed to be convex. Consequently, the power assignment and rate-splitting are jointly convex and can be solved within one subproblem by the MM algorithm.

D. Convergence Analysis of the BCD-Based Algorithm

The proposed BCD-based algorithm alternately solves the above three subproblems until the MACTR variation between consecutive iterations falls below the convergence threshold, as specified in **Algorithm 1**.

The convergence of the proposed algorithm follows from three key properties. First, the problem **(P0-1)** admits an upper-bounded optimal solution due to its compact feasible set and continuity. Second, for both subproblems in trajectory design, the introduction of auxiliary variables yields an equivalent reformulation that maintains the optimal value, despite increasing the dimensionality of the feasible set. Subsequent constraint approximations transform the original feasible set into its convex subset, whose optimal solution provides an LB to the original problem. Moreover, the IPM ensures that this LB never decreases after each iteration. Third, the final projection onto the feasible set of problem **(P0-1)** maintains the objective value. Unlike the subproblems in trajectory design, the resource allocation subproblem does not involve dimensionality expansion or projection operations. Consequently, the proposed BCD-based algorithm guarantees monotonically non-decreasing objective values throughout the optimization process and thus converges.

E. Computational Complexity Analysis

The primary computational complexity of horizontal coordinate arrangement arises from the IPM. Given the variable dimension $(4 + 3K + 2W)I + 1$ in the problem **(P1-1)** and an IPM convergence threshold η_1 , the complexity of IPM in one iteration is $\mathcal{O}(((4 + 3K + 2W)I + 1)^{3.5} \log_2(1/\eta_1))$ [49]. With $\mathcal{O}(1/\epsilon_1)$ iterations required to meet the convergence threshold ϵ_1 of SCA method [50], the computational complexity is $\varpi_1 = \mathcal{O}(((4 + 3K + 2W)I + 1)^{3.5} \log_2(1/\eta_1)/\epsilon_1)$.

Similarly, the computational complexity of altitude adjustment is $\varpi_2 = \mathcal{O}(((2 + 3K + 2W)I + 1)^{3.5} \log_2(1/\eta_2)/\epsilon_2)$,

where η_2 and ϵ_2 are the convergence thresholds of the IPM and GP-based approach, respectively.

For resource allocation, the computational complexities of updating coefficients and executing the IPM in each iteration are $\mathcal{O}(4KI)$ and $\mathcal{O}((2(K + 1)I + 1)^{3.5} \log_2(1/\eta_3))$, respectively, where η_3 is the convergence threshold of the IPM. Assuming the convergence threshold of the MM algorithm is ϵ_3 , its computational complexity is $\varpi_3 = \mathcal{O}(((2(K + 1)I + 1)^{3.5} \log_2(1/\eta_3) + 4KI)/\epsilon_3)$.

Algorithm 1 alternates between the above three algorithms, requiring $\mathcal{O}(1/\epsilon)$ iterations [51]. Consequently, its overall computational complexity is $\mathcal{O}((\varpi_1 + \varpi_2 + \varpi_3)/\epsilon)$. Given that the overall algorithm has polynomial time computational complexity, it can achieve high computational efficiency through joint software and hardware optimization strategies [52].

Moreover, in scenarios that are computationally constrained or have stringent latency requirements, the optimization objective can be approximated. For instance, the ratio of the user's received signal energy to the warden's received signal energy has been considered as the objective function [8]. Furthermore, the SNR has also been adopted as an approximation for the data rate [53]. Following a similar approach, the SINR is employed in this work to approximate the CTR, and the performance of this approximation is analyzed in Section IV-B.

VI. SIMULATION RESULTS

This section evaluates the proposed algorithm for RSMA-based VLCC in UAV systems with jammer assistance. The convergence behavior and key performance metrics are analyzed in Section VI-A. Moreover, comparative analyses against baseline methods and alternative MA schemes are conducted in Section VI-B, along with the impacts of critical parameters.

The system consists of 6 ground users at $(30 \text{ m}, 75 \text{ m})^T$, $(75 \text{ m}, 75 \text{ m})^T$, $(105 \text{ m}, 45 \text{ m})^T$, $(135 \text{ m}, 15 \text{ m})^T$, $(165 \text{ m}, 30 \text{ m})^T$, and $(180 \text{ m}, 60 \text{ m})^T$, along with 3 terrestrial wardens at $(60 \text{ m}, 60 \text{ m})^T$, $(135 \text{ m}, 45 \text{ m})^T$, $(180 \text{ m}, 75 \text{ m})^T$. Besides, UAVs S and J are initialized at $\mathbf{q}_{S,I} = (0 \text{ m}, 45 \text{ m})^T$ and $\mathbf{q}_{J,I} = (0 \text{ m}, 120 \text{ m})^T$. Their corresponding final positions are $\mathbf{q}_{S,F} = (210 \text{ m}, 45 \text{ m})^T$ and $\mathbf{q}_{J,F} = (210 \text{ m}, 120 \text{ m})^T$. Moreover, the initial and final altitudes for both UAVs are set to $z_{v,I} = z_{v,F} = 24 \text{ m}$. Other parameters remain unchanged as specified in Table III unless otherwise stated, following the configurations in [31] and [54].

A. Convergence and Key Performance Metrics

The convergence behavior of the proposed algorithm under different covertness constraints is illustrated in Fig. 2. The

$$\frac{1}{I} \sum_{i=1}^I \left(\frac{B}{2} \left(\alpha_k[i] \left(\log_2 \left(\frac{e}{2\pi} h_{S,k}^2[i] P_k[i] \right) - \log_2(\Omega_i(\hat{\mathbf{P}})) + \frac{\log_2 e}{\Omega_i(\hat{\mathbf{P}})} h_{S,k}^2[i] \sum_{l \in \mathcal{K}_k} (P_l[i] - \hat{P}_l[i]) \right) + \beta_k[i] \right) + C_k[i] \right) \geq \chi. \quad (70)$$

$$\frac{B}{2} \left(\alpha_{C,k}[i] \left(\log_2 \left(\frac{e}{2\pi} h_{S,k}^2[i] P_C[i] \right) - \log_2(\Omega_{C,i}(\hat{\mathbf{P}})) + \frac{\log_2 e}{\Omega_{C,i}(\hat{\mathbf{P}})} h_{S,k}^2[i] \sum_{l \in \mathcal{K}} (P_l[i] - \hat{P}_l[i]) \right) + \beta_{C,k}[i] \right) \geq \varsigma[i]. \quad (71)$$

Algorithm 1 BCD-Based MACTR Maximization Algorithm

- 1: **Input:** $Q_v^{(0)}, z_v^{(0)}, P^{(0)}, C^{(0)}, u_j$, and the threshold ϵ ;
- 2: **Output:** Q^*, z^*, P^* and C^* ;
- 3: **Init:** $r = 0, \varsigma^{(0)}[i] = \sum_{k \in \mathcal{K}} C_k^{(0)}[i]$, and $\chi^{(0)}$;
- 4: **Repeat**
- 5: $r = r + 1$;
- 6: Given $z^{(r-1)}, P^{(r-1)}, C^{(r-1)}$, and $\varsigma^{(r-1)}$, obtain $Q^{(r)}$ by the SCA method;
- 7: Based on $Q^{(r)}, P^{(r-1)}, C^{(r-1)}$, and $\varsigma^{(r-1)}$, calculate $z^{(r)}$ via the GP-based approach;
- 8: With $Q^{(r)}$ and $z^{(r)}$, optimize $P^{(r)}, C^{(r)}$, and $\varsigma^{(r)}$ through the MM algorithm;
- 9: **Until** $|\chi^{(r)} - \chi^{(r-1)}| < \epsilon$
- 10: $Q^* = Q^{(r)}, z^* = z^{(r)}, P^* = P^{(r)}, C^* = C^{(r)}$.

TABLE III
SIMULATION PARAMETERS

Parameter	Value
Detector area, A	1 cm ²
Lambert order, m	1
Refractive index, n_r	1.5
Optical filter gain, Υ	1
Field-of-view semi-angle, Ψ_c	60°
Bandwidth, B	80 MHz
Noise power, σ^2	-110 dBm
Flight duration, T	60 s
Time slot duration, δ	1 s
Covert constraint, ϵ	0.1
Horizontal distance constraint, $r_{\min}$	5 m
Maximum flight speeds, $V_{H,\max}, V_{A,\max}$	10 m/s, 4 m/s
Altitude constraints, $z_{\min}, z_{\max}$	18 m, 30 m
Maximum power constraints, $P_{S,\max}, P_{J,\max}$	10 W, 10 W
Average power constraint, $\bar{P}_{\max}$	2.5 W

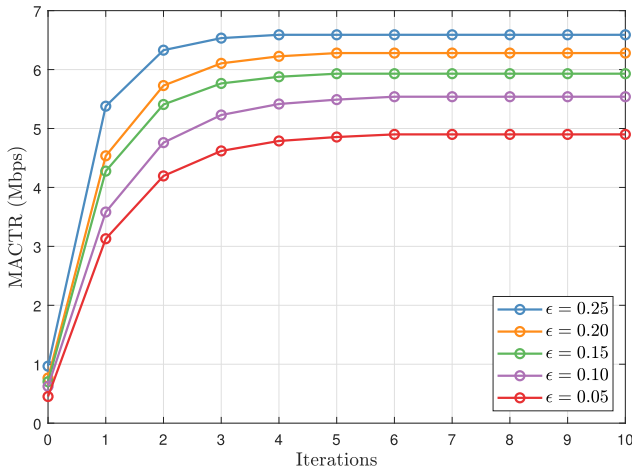


Fig. 2. Convergence of the proposed algorithm under varying covertness constraints.

MACTR increases monotonically with iterations and converges within merely 6 iterations. Compared to the initial values, the proposed algorithm yields a significant performance improvement. Taking $\epsilon = 0.10$ as an example, the MACTR increases substantially from 0.62 Mbps to 5.54 Mbps, highlighting the benefits of joint trajectory design and

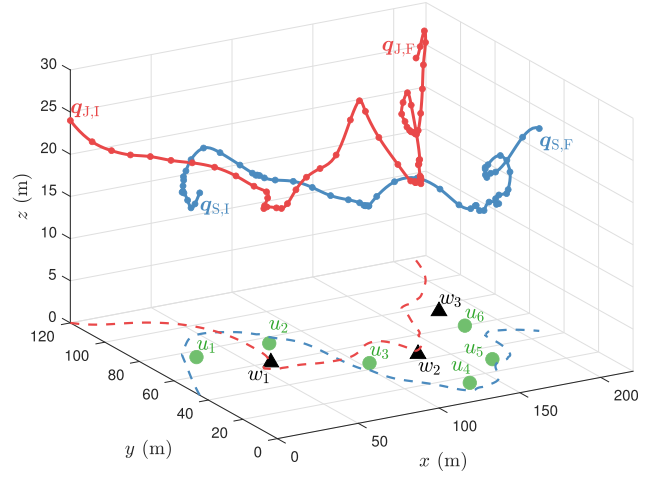


Fig. 3. Optimized 3D trajectories of source and jamming UAVs.

resource allocation. Furthermore, as the covertness requirement becomes more rigorous, characterized by a declining ϵ , the MACTR exhibits a decreasing trend, since UAV S reduces the transmit power to prevent the detection by wardens.

Fig. 3 demonstrates the optimized 3D trajectories of both UAVs, where the blue and red solid lines represent the trajectories of UAV S and UAV J, and the blue and red dashed lines represent their ground projections, respectively. In addition, users are represented by green solid circles and wardens by black triangles. Both are numbered in the figure, denoted as u_j or w_j for convenience, where j indicates their index. Notably, instead of following direct routes from initial to final positions, both UAVs sequentially visit users or wardens. Taking UAV S as an example, it exhibits adaptive velocity control, moving rapidly between users while decelerating to near-hovering states when approaching them, as indicated by denser trajectory points near solid green circles. This strategy enhances channel gains, thereby improving the MACTR. Moreover, UAV J simultaneously hovers near the warden closest to the user served by UAV S, leading to effective jamming. Additionally, the covertness constraint suggests that directly deploying UAV S above users is suboptimal. This is particularly evident when UAV S maintains a certain distance from user 6 due to its 15 m proximity to warden 3, preventing potential covert communication failure. Besides, the altitude selection of UAV S depends on the spatial distribution of all served users. For instance, UAV S raises its altitude when moving between users, while also maintaining elevated positions to better serve users 3, 4, and 5. Similarly, UAV J descends for precisely jamming near wardens and ascends to persistently jam distant ones. Furthermore, the proximity to adjacent users also influences the altitude optimization. Specifically, UAV J reaches a peak altitude of 29.6 m to reduce the interference with user 3.

Fig. 4 presents the transmit power of UAV S $P_S[i]$, CTR of UAV S $R_S[i]$, and DEPs of wardens $\xi_w[i]$ per time slot. The plateaus in both $P_S[i]$ and $R_S[i]$ exhibit temporal consistency, coinciding with periods when UAV S hovers in proximity to different users. Specifically, higher $P_S[i]$ values are observed during time slots 15 ~ 20 and 49 ~ 53, cor-

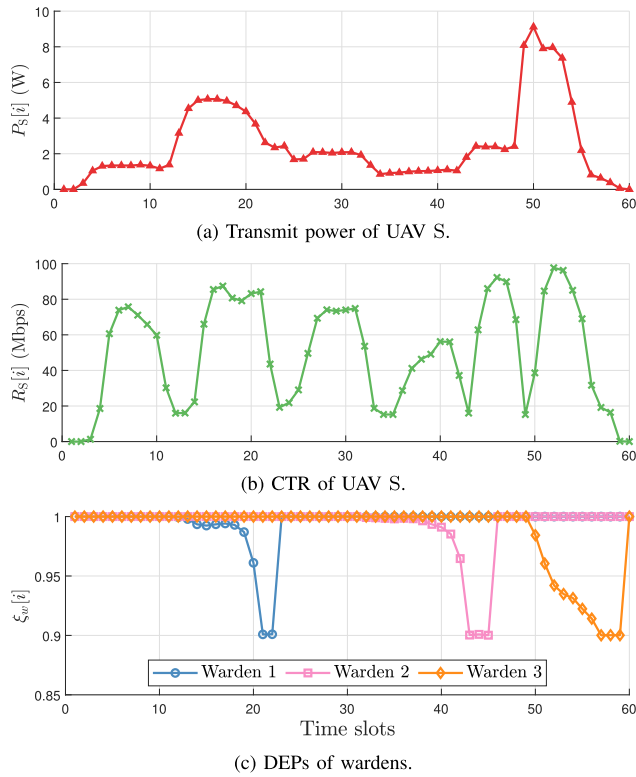


Fig. 4. Key performance metrics per time slot for the proposed VLCC system.

responding to service periods for users 2 and 6, respectively. This assignment pattern stems from the low SINR of user 2 and the inferior channel gain of user 6, requiring increased power to enhance the MACTR. Moreover, $R_S[i]$ experiences temporary reductions around time slots 12, 23, 35, 43, and 49 due to decreased channel gains during the movement of UAV S between users. Regarding $\xi_w[i]$, all time slots satisfy the covertness requirement, with most values approaching 1.0, demonstrating the advantage of the proposed algorithm. Additionally, the DEP curve transitions abruptly to 1.0 at time slots 23, 46, and 60, reflecting instances where wardens fail to detect signals due to the limited coverage of VLC. However, near-threshold DEP values occur at time slots 21, 22, 43 ~ 45, and 57 ~ 59, primarily because the increased distances from UAV J to wardens reduce the jamming effect.

B. Performance of Different Schemes and Parameters

To validate the effectiveness of the proposed algorithm and its SINR approximation variant, the following benchmarks are employed to solve individual subproblems while preserving the original optimization framework for the rest.

- 1) **Particle Swarm Optimization (PSO) for horizontal coordinate arrangement [30]**: This approach treats all possible UAV trajectories as particles and utilizes a memory-enhanced technique to iteratively find the optimal position in each time slot.
- 2) **Mutual Information Maximization Algorithm (MIMA) for altitude adjustment [26]**: Adapted from a single-user and non-jammer scenario, this algorithm only optimizes the altitude of UAV S to maximize

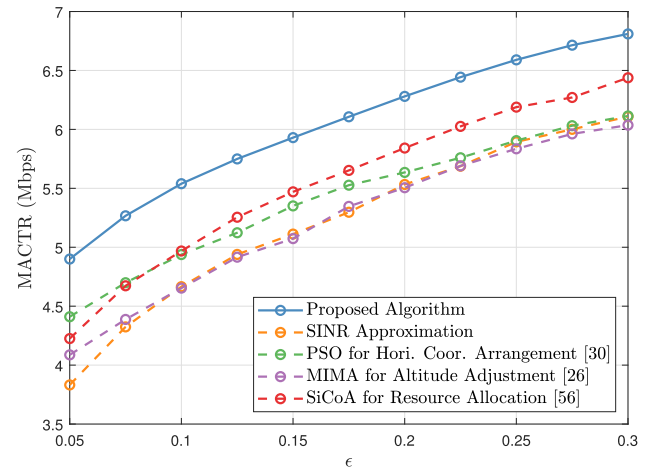


Fig. 5. MACTR performance versus the covertness constraint compared to different benchmarks.

the mutual information for the nearest user, while the altitude of UAV J remains fixed at $z_{J,I} = 24$ m.

- 3) **Sine-Cosine Algorithm (SiCoA) for resource allocation [55]**: This population-based metaheuristic algorithm is employed to optimize the transmit power and rate-splitting.

As shown in Fig. 5, all schemes demonstrate increasing MACTR with the relaxation of covertness constraint ϵ , and the proposed algorithm achieves superior performance. Specifically, at $\epsilon = 0.1$, the proposed algorithm outperforms the second-best scheme by 0.57 Mbps. This performance advantage stems from the fact that the proposed algorithm is specifically tailored to the investigated scenario, unlike the more general heuristic approaches of PSO and SiCoA. Moreover, the MIMA benchmark, originally designed for a single-user context, inherently incurs a performance loss when adapted to the multi-user access problem. Besides, it is noted that the approximation algorithm achieves approximately 85% of the MACTR obtained with the fully optimized proposed method, representing a deliberate trade-off for reduced computational complexity.

Additionally, the proposed RSMA scheme is compared with an existing SCA-based RSMA scheme, along with conventional NOMA, time-division multiple access (TDMA), and orthogonal frequency division multiple access (OFDMA) schemes, which are implemented as follows.

- 1) **SCA for RSMA [10]**: The SCA framework leverages the first-order Taylor expansion to transform the non-convex data rate maximization problem into iterative convex subproblems for power assignment.
- 2) **PSO for NOMA [56]**: This algorithm optimizes the power assignment by simulating swarm intelligence, where particles leverage both individual experience and social information sharing to guide the search process.
- 3) **Lagrangian Relaxation (LR) for OFDMA [57]**: The subchannel assignment variables are relaxed, and the non-convex constraint is weighted by Lagrangian multipliers as a penalty term, which are then iteratively updated employing the subgradient method.

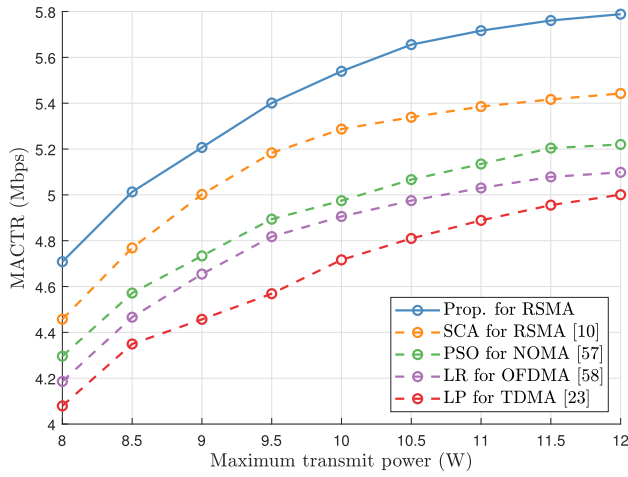


Fig. 6. MACTR performance comparison across MA schemes with respect to maximum transmit power.

4) Linear Programming (LP) for TDMA [23]: Through the relaxation technique, the LP approach is employed for optimal time slot allocation in TDMA scheme.

As illustrated in Fig. 6, the proposed RSMA scheme achieves the best MACTR across various maximum transmit power levels $P_{S,\max}$. Specifically, at $P_{S,\max} = 10$ W, it demonstrates MACTR gains of 0.25 Mbps, 0.57 Mbps, 0.63 Mbps, and 0.82 Mbps over the baseline RSMA, NOMA, OFDMA, and TDMA schemes, respectively. This performance gap can be attributed to the inherent limitations of the conventional schemes. For OMA schemes like OFDMA and TDMA, the strict orthogonality of the resource blocks leads to suboptimal resource utilization. For NOMA, its effectiveness is weakened by the dynamic SINR fluctuations caused by UAV mobility. These results validate the advantage of the proposed MM algorithm compared to the baseline SCA approach, while highlighting the superiority of RSMA over conventional MA schemes. Furthermore, while MACTR increases with $P_{S,\max}$ for all MA schemes, the growth rate gradually diminishes. This phenomenon occurs because the covertness constraint limits the full exploitation of additional transmit power for MACTR enhancement.

Fig. 7 demonstrates the relationship between MACTR and maximum jamming power $P_{J,\max}$ under different covertness constraints. A tradeoff is observed where MACTR first increases and subsequently decreases with escalating $P_{J,\max}$. This phenomenon results from dual effects of $P_{J,\max}$, i.e., insufficient $P_{J,\max}$ may fail to meet covertness requirements, while excessive $P_{J,\max}$ deteriorates the SINR of users. Moreover, the optimal $P_{J,\max}$ corresponding to the MACTR peak shifts to higher values as covertness constraints become more rigorous. This trend suggests that under strict covertness requirements, insufficient $P_{J,\max}$ restricts the full utilization of the transmit power of UAV S, thereby limiting the maximum MACTR, consistent with the findings presented in Fig. 6.

A comparison of MACTR for dynamic and static deployment schemes with varying numbers of UAVs is presented in Fig. 8. In this comparison, dynamic deployment corresponds to the proposed trajectory optimization scheme. For the static deployment case, the source UAV remains at a fixed position

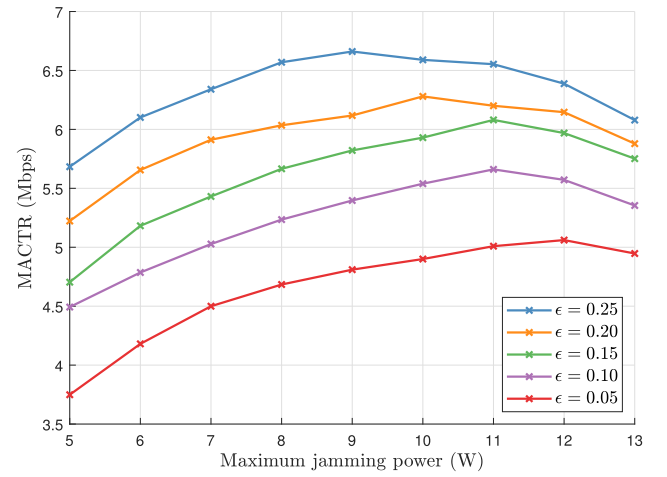


Fig. 7. MACTR performance versus the maximum jamming power under various covert constraints.

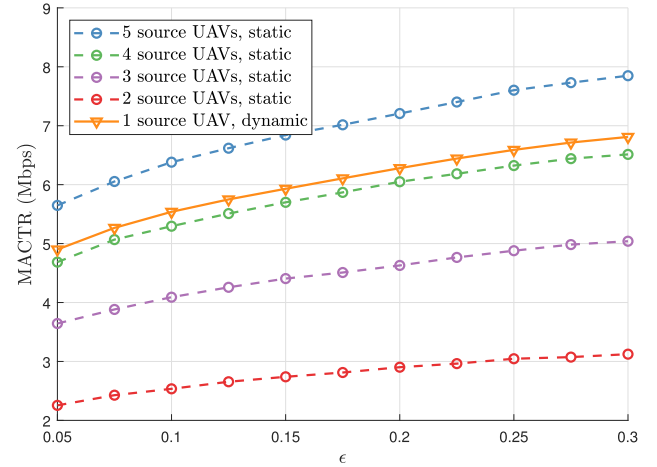


Fig. 8. Comparison of MACTR for dynamic and static deployment schemes with different UAV numbers.

throughout all time slots, while the jamming UAV still employs the proposed trajectory optimization algorithm to maintain covert communications. A notable observation is that the performance of a single UAV under the dynamic deployment scheme can even exceed that of four UAVs under the static scheme. This significant advantage is attributed to the fact that dynamic deployment allows the UAV to move closer to different users in various time slots, thereby maximizing the average channel gain. Therefore, the trajectory optimization is crucial for enhancing the MACTR for users distributed over a wide area.

Fig. 9 presents the MACTR performance versus the covertness constraint ϵ under different flight durations T . The results reveal that the MACTR increases with longer flight durations. Specifically, at $\epsilon = 0.1$, the MACTR for $T = 80$ s exceeds that for $T = 60$ s by 0.58 Mbps. This improvement stems from extended hovering periods near users as flight duration increases, thereby enhancing the proportion of time slots with strong channel gains. However, it is important to note that longer flight durations result in higher power consumption, requiring careful selection of the flight duration based on battery capacity and communication requirements.

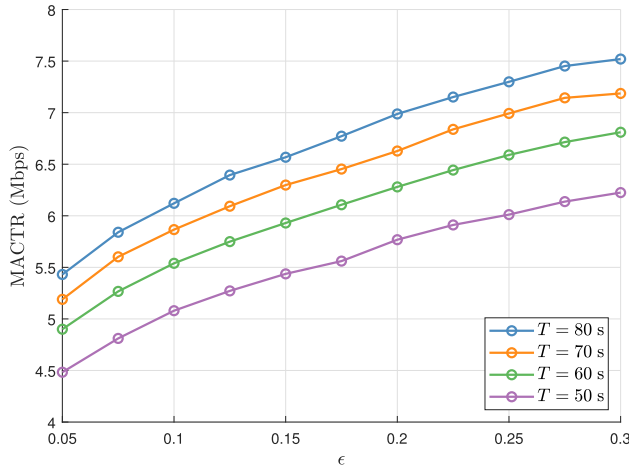


Fig. 9. MACTR performance versus the covertness constraint under different flight durations.

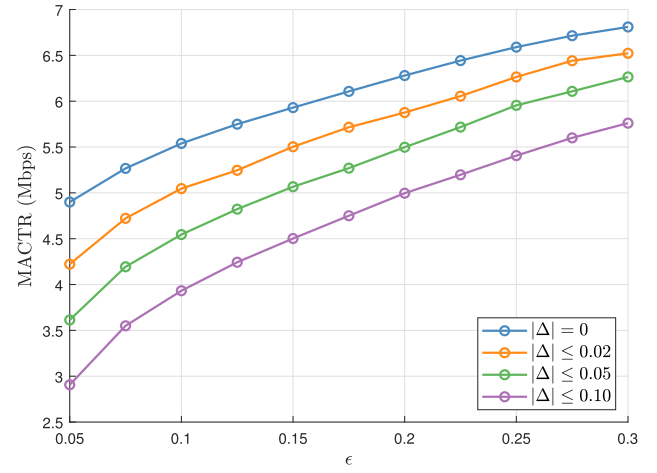


Fig. 11. MACTR versus the covertness requirement for different levels of CSI uncertainty.

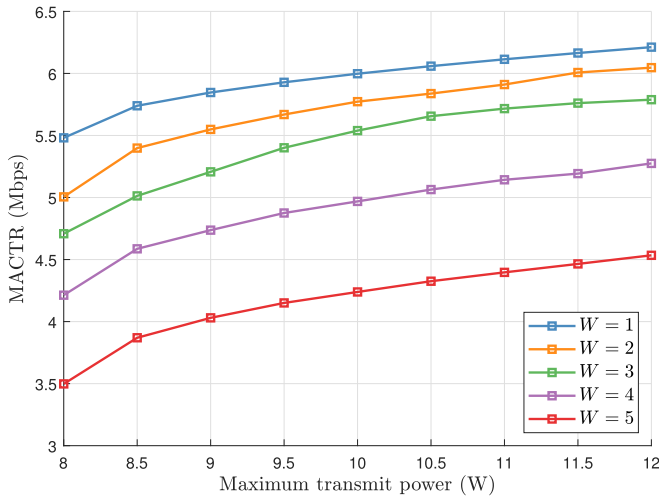


Fig. 10. MACTR versus the maximum transmit power for different warden numbers.

Fig. 10 shows the MACTR versus the maximum transmit power for a varying number of wardens. For this analysis, wardens with the lowest W indices are selected, and the locations of warden 4 and warden 5 are set to $(75 \text{ m}, 30 \text{ m})^T$ and $(180 \text{ m}, 20 \text{ m})^T$, respectively. The results indicate that a larger number of wardens leads to a lower MACTR. This is because the UAV trajectory optimization becomes more constrained as it must avoid proximity to more wardens, leading to a degradation in the average channel gain to the legitimate users. Specifically, at a maximum transmit power of 10 W, the MACTR for $W = 1$ is 1.76 Mbps higher than that for $W = 5$. Additionally, the trend of MACTR with respect to the maximum transmit power is observed to be consistent with the results shown in Fig. 6.

Acquiring accurate CSI from wardens is considered challenging, and the impact of imperfect CSI is illustrated in Fig. 11. Specifically, a bounded error CSI model is adopted [8], [38], where the estimated channel gain $\tilde{h}_{v,w}[i]$ is modeled as $\tilde{h}_{v,w}[i] = h_{v,w}[i](1 + \Delta)$ with Δ denoting the bounded uncertainty. To ensure the covertness constraint is

met, the optimization considers the worst-case channel estimation within the bound. As shown in Fig. 11, the MACTR decreases as the CSI uncertainty $|\Delta|$ increases. This is because a larger uncertainty forces a more conservative transmit power allocation for UAV S to ensure the covertness constraint is not violated. Furthermore, this performance degradation is more pronounced under stricter covertness requirements, as indicated by smaller values of ϵ , highlighting that accurate warden CSI is particularly critical in such scenarios.

VII. CONCLUSION

In this paper, the RSMA-based VLCC system with dual UAVs was investigated, where a source UAV transmitted confidential data to users while a jamming UAV obscured warden detection. The core objective was to maximize the MACTR by joint 3D trajectory design and resource allocation. To solve this non-convex problem, a BCD-based algorithm was proposed to iteratively address decomposed subproblems for horizontal coordinate arrangement, altitude adjustment, along with power assignment and rate-splitting. Simulation results demonstrated the effectiveness of the proposed algorithm by evaluating its performance under various parameters and showcasing its superiority over baseline methods and traditional MA schemes. The key findings revealed the critical role of UAV coordination in enhancing the MACTR while enabling effective jamming towards wardens. Furthermore, a fundamental tradeoff concerning jamming power was identified, where an optimal balance must be struck to ensure covertness without excessively degrading the SINR of legitimate users. In summary, this study highlights the potential of VLC-enabled UAV systems for covert communications.

APPENDIX A PROOF OF THEOREM 1

For simplicity, time slot indices are omitted in this section when no ambiguity arises. Since it cannot be determined whether $\Xi_{w,1} < \Xi_{w,2}$, the DEP ξ_w should be analyzed for two cases in the following.

Case 1: When $\Xi_{w,1} < \Xi_{w,2}$, ξ_w is given as

$$\xi_w = \begin{cases} 1, & \tau_w < \sigma_w^2, \\ 1 - \frac{\tau_w - \sigma_w^2}{\Xi_{w,1}}, & \sigma_w^2 \leq \tau_w \leq \Xi_{w,1} + \sigma_w^2, \\ 0, & \Xi_{w,1} + \sigma_w^2 < \tau_w < \Xi_{w,2} + \sigma_w^2, \\ \frac{\tau_w - \sigma_w^2 - \Xi_{w,2}}{\Xi_{w,1}}, & \Xi_{w,2} + \sigma_w^2 \leq \tau_w \leq \Xi_{w,1} + \Xi_{w,2} + \sigma_w^2, \\ 1, & \tau_w > \Xi_{w,1} + \Xi_{w,2} + \sigma_w^2. \end{cases} \quad (75)$$

Obviously, the minimum DEP ξ_w^* is 0 and the corresponding optimal detection threshold is $\tau_w^* \in [\Xi_{w,1} + \sigma_w^2, \Xi_{w,2} + \sigma_w^2]$.

Case 2: When $\Xi_{w,1} \geq \Xi_{w,2}$, we can have

$$\xi_w = \begin{cases} 1, & \tau_w < \sigma_w^2, \\ 1 - \frac{\tau_w - \sigma_w^2}{\Xi_{w,1}}, & \sigma_w^2 \leq \tau_w \leq \Xi_{w,2} + \sigma_w^2, \\ 1 - \frac{\Xi_{w,2}}{\Xi_{w,1}}, & \Xi_{w,2} + \sigma_w^2 < \tau_w < \Xi_{w,1} + \sigma_w^2, \\ \frac{\tau_w - \sigma_w^2 - \Xi_{w,2}}{\Xi_{w,1}}, & \Xi_{w,1} + \sigma_w^2 \leq \tau_w \leq \Xi_{w,1} + \Xi_{w,2} + \sigma_w^2, \\ 1, & \tau_w > \Xi_{w,1} + \Xi_{w,2} + \sigma_w^2. \end{cases} \quad (76)$$

Since $\xi_w = 1$ indicates the worst detection result, warden w does not set $\tau_w < \sigma_w^2$ or $\tau_w > \Xi_{w,1} + \Xi_{w,2} + \sigma_w^2$. Besides, ξ_w decreases monotonically with the increase in τ_w for $\sigma_w^2 \leq \tau_w \leq \Xi_{w,2} + \sigma_w^2$. In contrast, ξ_w is a monotonically increasing function of τ_w when $\Xi_{w,1} + \sigma_w^2 \leq \tau_w \leq \Xi_{w,1} + \Xi_{w,2} + \sigma_w^2$. Moreover, ξ_w remains constant outside these bounds and is a continuous function of τ_w . Therefore, the optimal detection threshold is $\tau_w^* \in [\Xi_{w,2} + \sigma_w^2, \Xi_{w,1} + \sigma_w^2]$ and the minimum DEP is $\xi_w^* = 1 - \Xi_{w,2}/\Xi_{w,1}$, which completes the proof. ■

APPENDIX B PROOF OF THEOREM 3

First, to address the SINR terms in (24a) and (24b), both the numerator and denominator are divided by $h_{S,k}^2[i]$. This manipulation recasts both terms into a similar structure, as exemplified by

$$\frac{\frac{e}{2\pi} P_C[i]}{(h_{J,k}^2[i] P_{J,\max}/2 + \sigma_k^2)/h_{S,k}^2[i] + \sum_{l \in \mathcal{K}} P_l[i]}, \quad (77)$$

in which only $h_{S,k}^2[i]$ and $h_{J,k}^2[i]$ are functions of the optimization variables. By introducing the auxiliary variable $\zeta_{k,H}[i]$, constraints (24a) and (24b) can be equivalently substituted by

$$\frac{h_{S,k}^2[i]}{\zeta_{k,H}[i]} \geq \frac{h_{J,k}^2[i] P_{J,\max}}{2} + \sigma_k^2, \quad (78)$$

together with (26) and (27). This equivalence holds because, when constraint (78) is imposed, the left-hand sides (LHSs) of (26) and (27) provide tight LBs for the LHSs of (24a) and (24b). Besides, these three constraints hold with equality at the optimal solution to problem (P1); otherwise, $\zeta_{k,H}[i]$ could

be increased to enlarge the LHSs of (26) and (27), thereby improving the objective value of problem (P1).

Nevertheless, constraint (78) remains computationally challenging. To address this, auxiliary variables $\vartheta_{k,H}[i]$ and $t_{k,H}[i]$ are further introduced. Specifically, when $\vartheta_{k,H}[i] \leq h_{S,k}[i]$, the term $\vartheta_{k,H}^2[i]/\zeta_{k,H}[i]$ is an LB for the LHS of (78). Similarly, $t_{k,H}[i] \geq h_{J,k}^2[i] P_{J,\max}/2$ serves as an upper bound for the first term of the right-hand side in (78). Moreover, the constraints associated with $\vartheta_{k,H}[i]$ and $t_{k,H}[i]$ can be rewritten as (29) and (30) via algebraic transformation. Consequently, an equivalent reformulation of (78) is given by (28), (29) and (30). This equivalence holds because if any of these three inequalities is not satisfied with equality, $\vartheta_{k,H}[i]$ can be increased or $t_{k,H}[i]$ can be decreased to enlarge $\zeta_{k,H}[i]$, thereby improving the objective value of problem (P1). Therefore, constraints (24a) and (24b) are substituted by (26)~(30).

Second, constraint (24d) is handled in a similar manner by introducing auxiliary variables $\iota_{w,H}^2[i] \geq h_{S,w}[i]$ and $\kappa_{w,H}[i] \leq h_{J,w}[i]$, which lead to the equivalent reformulation given by (31), (32), and (33). Since the transformation procedure closely parallels that of (78), further details are omitted for brevity. ■

APPENDIX C PROOF OF THEOREM 1

Consider the function $f(x) = \log_2(1 + A/(x^{-1} + B))$ defined for $x > 0$ with $A, B > 0$, which shares the convexity properties of the LHS of (27). The second derivative of $f(x)$ is calculated as

$$\frac{\partial^2 f}{\partial x^2} = -\frac{A}{\ln 2} \frac{2B(A+B)x + A + 2B}{((1+Bx)^2 + Ax(1+Bx))^2} < 0, \quad (79)$$

indicating the LHS of (27) is concave.

For the LHS of (28), the convexity follows directly from established results in [58, p. 73].

The term $d_{S,k}^p[i]$ with $p > 1$ can be expressed as a composition function $h \circ g$, where $h(x) = x^p$ is convex and non-decreasing while g denotes a convex norm. By the composition rules for convex functions [58, p. 84], $d_{S,k}^p[i]$ is convex. ■

APPENDIX D PROOF OF THEOREM 4

Following the derivation in Section V-A, constraints (24a) and (24b) and additional three constraints on auxiliary variables $\zeta_{k,A}[i]$, $\vartheta_{k,A}[i]$ and $t_{k,A}[i]$ can be equivalently transformed into

$$\frac{1}{\vartheta_{k,A}[i]} \geq \frac{1}{h_{S,k}[i]} = \frac{1}{h_0} \frac{d_{S,k}^{m+3}[i]}{z_S^{m+1}[i]}, \quad (80)$$

$$t_{k,A}[i] \geq \frac{P_{J,\max}}{2} h_{J,k}^2[i] = \frac{P_{J,\max} h_0^2}{2} \frac{z_J^{2m+2}[i]}{d_{J,k}^{2m+6}[i]}, \quad (81)$$

together with (43), (44), and (45). The right-hand sides of (80) and (81) possess monomial-like structures [47], which motivates the introduction of the variable substitution $z_v[i] = e^{-\kappa_v[i]}$. Applying this transformation and taking the logarithm of (80) and (81) yields (46) and (47). Therefore, constraints (24a) and (24b) are equivalently transformed into (43)~(47).

Besides, introducing auxiliary variables $\iota_{w,A}^2[i] \geq h_{S,w}[i]$ and $\kappa_{w,A}[i] \leq h_{J,w}[i]$ allows (24d) to be rewritten as (48), (49), and (50), which correspond to the logarithmic form of the results obtained in Section V-A. ■

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